

12. Programming Assignments (20%)

13. Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	B. Sc. Statistics				
Course Code	STA5FS112				
Course Title	Statistical analysis using Python				
Type of Course	SEC				
Semester	V				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	HSE level Mathematics Course				
Course Summary	To make students aware of Various Quality or standards in Industrial Production, Detecting, Controlling and Maintaining Quality and Total Quality Management				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain the basics of Python programming language and its syntax.	U	F	Instructor-created exams / Quiz
CO2	Identify common types of errors and their underlying causes such as incorrect syntax, invalid variable assignments, and type mismatches.	R	C	Instructor-created exams / Home Assignments/ Seminar Presentation

CO3	Describe the fundamental data structures in Pandas and their role in data manipulation and analyze data to help entrepreneurial decisions using critical thinking skills.	U	F	Instructor-created exams / Home Assignments/ Seminar Presentation
CO4	Familiarize themselves with popular data visualization libraries in Python and critically evaluate ethical implications of statistical methods aligning with human values.	R	C	Instructor-created exams / Quiz
CO5	Discuss the importance of random number generation in computational simulations and statistical analysis.	U	F	Instructor-created exams / Quiz/ Viva-Voce
CO6	Apply the basic concepts and principles of machine learning.	Ap	P	Instructor-created exams / Home Assignments/ Seminar Presentation
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Introduction to Python Programming		12	
	1	Interactive Python Environment: Jupyter notebooks, basic syntax, interactive shell	2	
	2	Data Fundamentals: Variables, assignments, arithmetic operators, expressions	3	
	3	Program Readability: Comments in code, interpreting error messages	3	
	4	Modular Programming: Importing modules, control flow statements	2	
	5	Function Fundamentals: Built-in and user-defined functions, arguments, return values, formal vs. actual parameters, named arguments	2	
	Sections from References: 2, 3, 5, 6			
II	Data Manipulation with Pandas		10	

	5	Pandas Introduction: Data Series, DataFrames	4	
	6	Data Operations: Importing, manipulating, merging, analyzing, and exporting DataFrames	4	
	7	Descriptive Statistics: Exploratory data analysis techniques	2	
	Sections from References: 1, 3, 7, 11			
III	Data Visualization		8	
	9	Data Visualization Libraries: Matplotlib, Seaborn, Plotly, ggplot, Geoplotlib, Pandas (and potentially others)	2	
	10	Plot-I : Line plot, bar plot, pie chart, box plot, histogram, strip plot, swarm plot,	3	
	11	Plot-II: Scatter plot, heatmap, density plot, cumulative frequencies, error bars	3	
	Sections from References: 1, 4, 9, 10			
IV	Statistical Data Analysis Using statsmodels		18	
	18	Random Number Generation	3	
	19	Correlation	2	
	20	Hypothesis Testing -I: One sample, two sample and paired t test	2	
	21	Hypothesis Testing -II: One way and Two way ANOVA	3	
	22	Hypothesis Testing -III: Non Parametric Tests	3	
		Linear Regression Modeling: Simple and multiple linear regression	3	
		Logistic Regression Models	2	
	Sections from References: 4, 9,10,11			
V	Open Ended Module:		12	
	1	Numerical Methods with NumPy: Efficient arrays and linear algebra operations	4	
	2	Machine Learning Introduction: Fundamentals of machine learning with scikit-learn	4	
	3	Web Data Scraping: Scraping web data using requests and BeautifulSoup	4	
	Sections from References: 7, 9, 11			

Books and References:

1. Embarak, D. O., Embarak, & Karkal. (2018). *Data analysis and visualization using python*. Berkeley, CA, USA: Apress.
2. Gowrishankar, S., & Veena, A. (2018). *Introduction to Python programming*. Chapman and Hall/CRC.
3. Guttag, J. V. (2016). *Introduction to computation and programming using Python: With application to understanding data*. MIT press.
4. Haslwanter, T. (2016). *An introduction to statistics with python. With Applications in the Life Sciences*; Springer International Publishing: Cham, Switzerland.
5. Lambert, K. A., & Osborne, M. (2015). *Fundamentals of PYTHON*. Cengage Learning, IE.
6. Lutz, M. (2013). *Learning python: Powerful object-oriented programming*. " O'Reilly Media, Inc."
7. McKinney, W. (2012). *Python for data analysis: Data wrangling with Pandas, NumPy, and IPython*. " O'Reilly Media, Inc."
8. Severance, C. (2016). *Python for everybody: Exploring Data using python 3*. Charles Severance.
9. Tattar, P., Ojeda, T., Murphy, S. P., Bengfort, B., & Dasgupta, A. (2017). *Practical Data Science Cookbook*. Packt Publishing Ltd.
10. Unpingco, J. (2016). *Python for probability, statistics, and machine learning*. Cham, Switzerland: Springer International Publishing.
11. VanderPlas, J. (2016). *Python data science handbook: Essential tools for working with data*. " O'Reilly Media, Inc."

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	-	-	-	3	3	-	-	-	-	-
CO 2	3	2	-	-	-	3	3	-	-	-	-	-
CO 3	-	-	2	-	-	2	1	-	-	3	3	-
CO 4	2	-	2	-	-	-	2	-	-	3	-	2
CO 5	-	-	2	1	2	-	2	-	1	3	-	-
CO 6	2	2	-	-	-	2	3	-	-	3	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

5. Quiz / Assignment/ Quiz/ Discussion / Seminar
6. Midterm Exam
7. Programming Assignments (20%)
8. Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal	Assignm	Project	End Semester
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